

NEURAL SYSTEMS

DEVELOPMENTAL SYSTEMS

Phenotype
under selection

Network
circuit or regulatory

Cells
identity

Molecules
channels, genes, enhancers

Stomatogastric
ganglion · crabs

Swim circuit
two nudibranchs

Cortex
human vs mouse

eve stripe 2
fruit fly species

Gap genes
two fly species

Vulva
two nematodes

pyloric rhythm

swimming

stripe 2 expression

gap expression pattern

vulval cell fates

synaptic strengths (models)

role of Si1 in the circuit

interaction strengths

EGF vs Wnt induction

identified neurons

homologous neurons

matched cell types

precursor cells

K⁺ conductances, 2–4×

gene expression

enhancer binding sites

maternal inputs

Ras activity (cryptic)

 conserved between individuals or species  varies or turns over  not examined